10TH CLASS PHYSICS — NEW SYLLABUS (PECTAA)

Chapter 20 — Atomic and Nuclear Physics

Solved Exercise | New Book (PECTAA 2026)

Exercise 20 (Solved)

This exercise covers the structure of the atom, Rutherford's model, the nucleus, isotopes, radioactivity, the three types of radiation, half-life, $E = mc^2$, nuclear fission and fusion, carbon dating and the effects and uses of nuclear radiation.

A. Multiple Choice Questions

20.1 What did Rutherford's alpha particle scattering experiment reveal about the structure of the atom?

(a) Electrons are distributed evenly throughout the atom
(b) Protons and electrons are located in the nucleus
(c) Atoms are indivisible and indestructible
(d) The nucleus is small, dense, and positively charged

Answer: (d) The nucleus is small, dense, and positively charged — most alpha particles passed straight through the foil, but a few were strongly deflected back — possible only if the positive charge and nearly all the mass are concentrated in a tiny central nucleus.

20.2 The atomic number of an element represents:

(a) number of neutrons in the nucleus
(b) number of protons in the nucleus
(c) total number of protons and neutrons
(d) number of electrons and neutrons

Answer: (b) number of protons in the nucleus — the atomic number Z is the number of protons (and hence, for a neutral atom, the number of electrons); the mass number A is protons + neutrons.

20.3 Which type of radiation is positively charged and consists of two protons and two neutrons?

(a) Alpha (α)
(b) Beta (β)
(c) Gamma (γ)
(d) Neutron emission

Answer: (a) Alpha (α) — an alpha particle is a helium nucleus $^{4}_{2}He$ — two protons and two neutrons, so it carries a charge of +2e.

20.4 The process by which an unstable nucleus emits radiation and transforms into a more stable nucleus is called:

(a) nuclear fusion
(b) nuclear fission
(c) radioactive decay
(d) isotopic exchange

Answer: (c) radioactive decay — an unstable (radioactive) nucleus spontaneously emits alpha, beta or gamma radiation, loses energy and becomes a more stable nucleus.

20.5 Nuclear fission involves:

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(a) combining light nuclei to form heavier ones
(b) splitting heavy atomic nuclei into smaller ones
(c) absorbing gamma radiation to increase stability
(d) none of the above

Answer: (b) splitting heavy atomic nuclei into smaller ones — in fission a heavy nucleus such as uranium-235 splits into two lighter nuclei with the release of energy; combining light nuclei is fusion.

20.6 What does Einstein's equation; $E = mc^2$; explain?

(a) The energy released in chemical reactions
(b) The equivalence of mass and energy
(c) The rate of radioactive decay
(d) The conversion of mass into momentum

Answer: (b) The equivalence of mass and energy — the equation shows that a mass m is equivalent to an amount of energy $E = mc^2$, so a small loss of mass in a nuclear reaction appears as a very large amount of energy.

20.7 Which of the following is a practical application of nuclear radiations?

(a) Cooking food in microwave ovens
(b) Medical imaging and cancer treatments
(c) Solar energy generation
(d) Airplane navigation systems

Answer: (b) Medical imaging and cancer treatments — X-rays, gamma rays and radioactive tracers are used in diagnosis, in radiotherapy and in sterilising equipment; the other three do not use nuclear radiation.

20.8 What are the potential effects of ionizing nuclear radiation on biological tissues?

(a) Increased cellular regeneration
(b) DNA damage and cancer
(c) Enhanced immune system response
(d) Improved metabolic activity

Answer: (b) DNA damage and cancer — ionizing radiation can break the chemical bonds in DNA, causing mutations, cell death, burns and, in the long term, cancer or genetic damage.

B. Short Answer Questions

20.1 Define the atomic nucleus. What are its two main characteristics?

The atomic nucleus is the small, heavy, central part of an atom, made up of protons and neutrons (together called nucleons), around which the electrons revolve. It carries the whole of the positive charge of the atom.

Its two main characteristics are: (i) it is extremely small — its radius is of the order of 10^{-15} m, about one hundred-thousandth of the size of the atom — yet it contains almost the entire mass of the atom; and (ii) it is positively charged, the charge being +Ze where Z is the number of protons. The protons and neutrons inside are held together by the very strong short-range nuclear force, and the number of protons decides the identity of the element.

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20.2 What is the structure of an atom according to Rutherford's model?

According to Rutherford's model, an atom consists of a very small, dense, positively charged central core called the nucleus, which contains almost all the mass of the atom. Around the nucleus, the light, negatively charged electrons revolve in circular orbits, in the same way as the planets revolve around the Sun.

The electrons are held in their orbits by the electrostatic force of attraction of the positive nucleus, exactly as gravity holds the planets. The atom as a whole is neutral, because the number of electrons equals the number of protons. The model also shows that the atom is mostly empty space: the nucleus is about 10^{-15} m across while the atom is about 10^{-10} m.

Rutherford's model could not explain the stability of the atom or the line spectra of hydrogen; these were explained later by Bohr, who allowed only certain fixed orbits with quantised energy.

20.3 How does alpha radiation differ from beta radiation?

Point	Alpha radiation (α)	Beta radiation (β)
Nature	a helium nucleus, $^{4}_{2}\mathrm{He}$ — two protons and two neutrons	a fast-moving electron (β^{-}) or a positron (β^{+})
Charge	+2e (positive)	−1e (negative) for β^{-}
Mass	about 4 u — heavy	about 1/1836 u — very light
Ionizing power	very strong (about 100 times that of beta)	weak, about a hundredth of alpha
Penetrating power	small — stopped by a sheet of paper or a few centimetres of air	greater — stopped by a few millimetres of aluminium
Speed	about 5 % of the speed of light	up to about 90 % of the speed of light
Effect on nucleus	mass number falls by 4 and atomic number by 2	mass number stays the same, atomic number rises by 1 (for β^{-})
Example	$^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$	$^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\mathrm{e}$

20.4 What is the primary energy source of the Sun and other stars?

The primary energy source of the Sun and the stars is nuclear fusion. In the extremely hot core of the Sun, at about 15 million kelvin, hydrogen nuclei (protons) fuse together to form helium nuclei. In the proton–proton cycle, four hydrogen nuclei combine to give one helium nucleus, with the release of two positrons, two neutrinos and a large amount of energy:

$$4\ {}^{1}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + 2\ \mathrm{e}^{+} + \text{energy (about 26.7 MeV)}$$

The mass of the helium nucleus is slightly less than the mass of the four protons, and this missing mass (the mass defect) appears as energy according to $E = mc^2$. About 4 million tonnes of mass is converted into energy every second in the Sun, and this energy keeps the star shining for billions of years. So the Sun is a huge natural fusion reactor.

20.5 Name the equation proposed by Einstein that relates energy and mass.

Einstein's mass–energy equation is $E = mc^2$, where E is the energy obtained, m is the mass converted and $c = 3 \times 10^8$ m s^{-1} is the speed of light.
Because c^2 is such a large number (9×10^{16} m^2 s^{-2}), a tiny mass corresponds to an enormous energy: 1 g of matter, if completely converted, would give 9×10^{13} J. This equation explains why nuclear reactions release far more energy than chemical reactions, and it is the basis of nuclear power, the Sun's energy and the atomic bomb.

20.6 What is the principle behind carbon dating and its applications?

Principle. Cosmic rays produce carbon-14 in the upper atmosphere, and living plants take it in through photosynthesis; animals get it by eating plants. So all living things contain a small, constant proportion of radioactive carbon-14 to ordinary carbon-12 — about one atom of C-14 for every 10^{12} atoms of C-12.
When the organism dies, the intake of carbon stops. The carbon-14 already present keeps decaying with its half-life of 5,730 years, while carbon-12 stays unchanged. By measuring the remaining proportion of C-14 in a sample (from its activity), and comparing it with that in a living sample, the number of half-lives that have passed — and hence the age — can be calculated: $N = N_0(1/2)^n$, with $n = t / T_{1/2}$.
Applications: dating wood, charcoal, bones, cloth, mummies, peat and archaeological artefacts up to about 50,000 years old; dating ancient settlements and tools; checking the authenticity of old paintings and manuscripts; and studying past climates and the history of the Earth. It is not useful for samples older than about 50,000 years, because by then too little C-14 is left to measure.

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20.7 Define nuclear fusion with an example.

Nuclear fusion is the process in which two light nuclei combine (fuse) to form a single heavier nucleus, with the release of a very large amount of energy. Fusion occurs only at extremely high temperatures and pressures, because the nuclei must be brought close enough together to overcome their mutual electrostatic repulsion.
Example: in the Sun, four hydrogen nuclei fuse to form one helium nucleus with the release of about 26.7 MeV of energy. In a future fusion reactor, deuterium and tritium would be fused:

$$^{2}_{1}H + ^{3}_{1}H \rightarrow ^{4}_{2}He + ^{1}_{0}n + 17.6\ MeV$$

The mass of the product is slightly less than the total mass of the reactants, and this mass defect appears as energy according to $E = mc^2$. Fusion releases more energy per kilogram of fuel than fission, and its fuel (hydrogen from water) is plentiful.

20.8 Americium-241 ($^{241}_{95}Am$) undergoes alpha decay and produces an alpha particle. Write down the chemical equation for it and name the nucleus formed.

In alpha decay the nucleus loses an alpha particle ($^{4}_{2}He$), so its mass number falls by 4 and its atomic number falls by 2. The decay equation is:

$$^{241}_{95}Am \rightarrow {}^{237}_{93}Np + {}^{4}_{2}He + \text{energy}$$

Check: mass numbers 241 = 237 + 4 ✓ and atomic numbers 95 = 93 + 2 ✓. The nucleus formed is neptunium-237 ($^{237}_{93}Np$). Americium-241 is used in very small quantities in smoke detectors: the alpha particles ionize the air and allow a small current to flow, and the current is reduced when smoke enters.

C. Constructed Response Questions

20.1 If an atom is mostly empty space, why do not we fall through solid objects or pass through walls?

It is true that the nucleus occupies only about one part in 10^{15} of the volume of an atom, so an atom is almost entirely empty space. But the 'empty' space around the nucleus is not truly empty: it is filled by the electron cloud, the region in which the electrons move. The electrons of one atom cannot simply pass into the space occupied by the electrons of another atom, because both are negatively charged and repel each other strongly.

More importantly, the electrons are governed by quantum mechanics and by the Pauli exclusion principle: no two electrons can occupy the same quantum state. When the atoms of your hand are pressed against the atoms of a wall, their electron clouds overlap, and the exclusion principle and the electrostatic repulsion produce a very strong resisting force. The atoms are pushed back rather than interpenetrating. So the apparent solidity of matter does not come from solid particles blocking the way but from the electric forces and quantum rules between the electron clouds. That is why we cannot push our hand through a wall, although the nuclei themselves occupy almost no space.

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20.2 Imagine a new radioactive element is discovered. How could scientists estimate its half-life without waiting thousands of years?

The half-life can be found from the rate of decay, because the activity of a sample is proportional to the number of undecayed nuclei present: $A = \lambda N$, where λ is the decay constant. The half-life is then $T_{1/2} = \ln 2 / \lambda = 0.693 / \lambda$.

Procedure: (i) take a sample of known mass and calculate the number of atoms N in it, using the molar mass and Avogadro's number; (ii) measure the activity A — the number of decays per second — with a Geiger–Müller counter or a scintillation detector, correcting for background radiation; (iii) calculate $\lambda = A / N$, and hence the half-life. Because the measurement takes only a few minutes or hours, a half-life of thousands of years can be found easily.

Alternatively, the activity (or the mass of the parent) is measured at regular intervals and a graph of activity against time is plotted. If the activity falls to a

fraction such as 1/2, 1/4 or 1/8, the corresponding time gives the half-life directly, and the graph can be extrapolated for very long half-lives. A check with a known formula, $N = N_0(1/2)^{t/T}$, on a short observation period confirms the result.

20.3 If gamma rays do not change the number of protons or neutrons in a nucleus, why are they still considered dangerous?

Gamma rays are electromagnetic waves of extremely short wavelength and very high energy photons (above about 10 keV). Although they carry no charge and no mass and therefore do not change the proton or neutron number of a nucleus, they are the most penetrating of the three radiations — they can pass through the body and through several centimetres of lead.

The danger comes from their ionizing power. As a gamma photon passes through living tissue it knocks electrons out of atoms and molecules, breaking chemical bonds. When this happens in the DNA of a cell, the genetic instructions are damaged: the cell may die, or it may divide out of control and become a cancer cell. Gamma radiation can also cause radiation burns, cataracts, sterility and, at high doses, acute radiation sickness and death. Because gamma rays penetrate so deeply, they can reach the internal organs, which alpha and beta particles cannot.

That is why gamma sources are shielded with thick lead or concrete, handled with long tongs and kept at a distance, and why workers wear film badges to monitor their exposure. In medicine the same property is used to advantage: gamma rays from cobalt-60 are used to destroy cancer cells in radiotherapy, the beam being aimed carefully so that healthy tissue receives as little dose as possible.

20.4 If carbon-14 stopped forming in the atmosphere today, how would that affect the use of carbon dating in the future?

Carbon dating works because living things keep exchanging carbon with the atmosphere, so the proportion of carbon-14 in their bodies stays equal to that in the air; after death the C-14 decays without being replaced, and the fall in its proportion gives the age. If no new C-14 were formed, that constant reference level in the atmosphere would disappear.

In the immediate future the method would still work for objects that died before the change, because their C-14 started decaying from the old equilibrium value. But as time went on, all living things would contain less and less C-14, and the 'living standard' used for comparison would keep falling. Two hundred years later a fresh sample would already contain a measurable deficit, and the calibration would be wrong unless it were corrected.

After several half-lives — say 30,000 to 50,000 years — essentially no C-14 would be left anywhere, so no sample could be dated at all and the method would become useless. Scientists would then have to fall back on other radioactive clocks, such as potassium-40/argon-40, uranium-lead or thermoluminescence dating. The change would therefore first distort and then finally end the use of carbon dating.

20.5 What is the difference between nuclear fission and nuclear fusion in terms of process and energy release?

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Point	Nuclear fission	Nuclear fusion
Process	a heavy nucleus (U-235, Pu-239) splits into two lighter nuclei, usually after absorbing a neutron	two light nuclei (H-2, H-3) combine to form one heavier nucleus
Conditions	room temperature; needs a critical mass and a neutron to start	extremely high temperature (millions of K) and high pressure
Products	two fission fragments, 2–3 neutrons and gamma rays	a heavier nucleus (He), a neutron and gamma rays
Energy per event	about 200 MeV per fission	about 17.6 MeV per fusion (but per kilogram it is several times larger)
Energy per unit mass	large	still larger — about 4 times that of fission
Radioactive waste	a large amount of long-lived radioactive waste	very little radioactive waste; the product helium is harmless
Fuel	uranium, which is limited and must be enriched	hydrogen isotopes from water — practically unlimited
Where it happens	nuclear power plants, atomic bomb	the Sun and stars, hydrogen bomb; fusion reactors are still experimental

In both processes the energy comes from the mass defect: the total mass of the products is slightly less than that of the reactants, and the missing mass appears as energy by $E = mc^2$.

D. Comprehensive Questions

20.1 How does the structure of an atom determine its chemical properties? Explain with reference to the arrangement of protons, neutrons, and electrons?

The chemical properties of an element are decided by the number and arrangement of its electrons, especially the electrons in the outermost shell (the valence electrons), because chemical reactions involve only the transfer or sharing of these outer electrons. The nucleus and the inner electrons take almost no part in ordinary chemistry.

Role of protons. The number of protons (the atomic number Z) fixes the number of electrons in the neutral atom and hence the electron configuration. For example, Z = 11 gives sodium, 2, 8, 1 — one loosely held outer electron, so sodium is a very reactive metal; Z = 17 gives chlorine, 2, 8, 7 — one electron short of a full shell, so chlorine is a reactive non-metal. The two combine to form sodium chloride. In this way the proton number decides the position of the element in the periodic table and

all its chemical behaviour.

Role of electrons. Atoms tend to reach a stable noble-gas configuration, so elements with one or two valence electrons lose them (metals) and those with six or seven gain them (non-metals). The number of shells decides the size of the atom and the shielding of the outer electrons, so reactivity changes down a group as well as across a period.

Role of neutrons. Neutrons carry no charge and do not affect the electron arrangement, so they do not change the chemical properties. They only add mass and hold the nucleus together; isotopes of the same element (same protons, different neutrons) have identical chemistry but different masses and different nuclear stability. If the nucleus becomes unstable, however, the atom may be radioactive and decay into another element.

20.2 Explain Einstein's equation, $E = mc^2$, and its significance. How does this equation relate to nuclear reactions?

Einstein's equation $E = mc^2$ states that mass and energy are two forms of the same thing and can be converted into each other; a mass m is equivalent to an energy $E = mc^2$, where $c = 3 \times 10^8$ m s^{-1}. The factor c^2 is enormous (9×10^{16}), so a very small mass corresponds to a very large energy.

Significance. It explains the source of the Sun's energy and of nuclear power; it shows why the mass of a stable nucleus is less than the sum of the masses of its separate protons and neutrons (the mass defect) — the missing mass is the binding energy released when the nucleus was formed. It also underlies the idea of nuclear energy, radiation, and even the annihilation of matter and antimatter, and it is the basis of all energy calculations in nuclear physics.

Relation to nuclear reactions. In both fission and fusion the total mass of the products is slightly less than the total mass of the reactants. This mass difference Δm is not lost; it appears as energy, $E = \Delta m\, c^2$. For example, in the fission of one uranium-235 nucleus about 0.2 u of mass disappears, giving about 200 MeV of energy; in the fusion of deuterium and tritium about 0.02 u disappears, giving 17.6 MeV. In a nuclear reactor 1 kg of uranium releases energy equivalent to burning about 3,000 tonnes of coal, which is possible only because of this conversion.

20.3 Describe the process of radioactive decay and how it leads to the emission of alpha, beta, and gamma radiation?

Radioactive decay is the spontaneous disintegration of an unstable nucleus, in which it emits a particle or a photon and changes into a more stable nucleus. The decay is random: we cannot say which particular nucleus will decay next, but for a large number of nuclei the rate is fixed and is measured by the half-life. Energy and momentum are conserved in every decay, and the emitted radiation carries the surplus energy.

Alpha decay. When a heavy nucleus has too many protons and neutrons, it ejects an alpha particle ($^4_2$He), i.e. two protons and two neutrons. The mass number falls by 4 and the atomic number by 2, so the element changes into another element two places to the left. Example: $^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^4_2He$. Alpha emission is common in

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heavy nuclei such as uranium, radium and americium.

Beta decay. When a nucleus has too many neutrons compared with protons, a neutron changes into a proton and an electron (plus an antineutrino), and the fast electron is emitted as a beta particle: ${}^{1}_{0}n \rightarrow {}^{1}_{1}p + {}^{0}_{-1}e$. The mass number stays the same and the atomic number rises by 1. Example: ${}^{14}_{6}C \rightarrow {}^{14}_{7}N + {}^{0}_{-1}e$. A positron may be emitted instead when there are too many protons.

Gamma emission. After alpha or beta decay the nucleus is often left in an excited state. It then loses the extra energy by emitting a high-energy photon (a gamma ray). Neither the mass number nor the atomic number changes, because no particle is emitted, only energy: example, ${}^{60}_{27}Co^{*} \rightarrow {}^{60}_{27}Co + \gamma$. In this way each type of radiation comes from a particular kind of instability inside the nucleus.

20.4 Why do different isotopes of the same element have similar chemical properties but different physical properties? Explain.

Isotopes are atoms of the same element with the same number of protons but different numbers of neutrons, so their mass numbers differ. For example, ${}^{12}_{6}C$, ${}^{13}_{6}C$ and ${}^{14}_{6}C$ all have six protons but six, seven and eight neutrons respectively.

Why the chemistry is the same. Chemical properties depend on the arrangement of the electrons, especially the outermost ones, and the number of electrons equals the number of protons. Since all isotopes have the same proton number, they have the same electron configuration. They therefore form the same compounds and take part in exactly the same reactions. The small difference of mass has only a very slight effect on reaction rates.

Why the physical properties differ. Neutrons add mass without adding charge, so the mass of the atom and of its nucleus changes. This changes properties such as density, melting and boiling point, rate of diffusion and the mass of the molecule. More importantly, the ratio of neutrons to protons decides the stability of the nucleus: some isotopes are stable while others are radioactive, with different half-lives and different types of radiation. That is why ${}^{12}_{6}C$ is stable, ${}^{14}_{6}C$ is radioactive and used in carbon dating, and ${}^{235}_{92}U$ is fissionable while ${}^{238}_{92}U$ is not. So isotopes are chemically alike but physically and nuclear-wise different.

20.5 Explain how carbon dating uses carbon-14 to estimate the age of artifacts. Discuss its reliability, considering the influence of carbon-14's half-life?

How it works. Carbon-14 is produced continuously in the upper atmosphere when cosmic-ray neutrons strike nitrogen-14 nuclei. It mixes with ordinary carbon dioxide, and plants absorb it during photosynthesis; animals take it in through food. So every living organism has the same small proportion of C-14 as the atmosphere (about 1 in 10^{12} carbon atoms), and this proportion stays constant while it lives.

At death the exchange stops. From then on the C-14 in the remains decays (half-life 5,730 years) into nitrogen-14, while carbon-12 stays unchanged. By measuring the remaining activity of the sample with a counter and comparing it with that of a fresh living sample, the number of half-lives n is found, and the age is $t = n \times 5{,}730$ years. For example, a sample with 25 % of the original C-14 has gone through two

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half-lives, so it is about 11,460 years old.

Reliability. It is very reliable for samples up to about 30,000–40,000 years old, and the method is calibrated with tree rings and other dating methods. Its limits come from the half-life: after 10 half-lives (57,000 years) only about 0.1 % of the C-14 is left, and the count becomes too small to measure accurately; so older objects cannot be dated this way. Contamination by newer carbon (from handling or from the soil) makes a sample look too young, and industrial burning of fossil fuels and nuclear tests have changed the atmospheric C-14 level in modern times. For these reasons the result is always quoted with an uncertainty and, where possible, is checked against another method.

E. Numerical Problems

20.1 Find the number of protons and neutrons in the nuclide defined by $^{21}_{9}X$.

Given: mass number A = 21
atomic number Z = 9

To find: protons = ?, neutrons = ?

Formula: protons = Z, neutrons = A − Z

Solution: protons = 9
neutrons = 21 − 9 = 12

Answer: 9 protons and 12 neutrons

20.2 A fossil is found to have 25% of the carbon-14 compared to a living sample. How old is the fossil?

Given: $N / N_0 = 25\ \% = 1/4$
$T_{1/2}$ of C-14 = 5,730 years

To find: t = ?

Formula: $N / N_0 = (1/2)^n$, $t = n \times T_{1/2}$

Solution: $(1/2)^n = 1/4 = (1/2)^2$, so n = 2 half-lives
$t = 2 \times 5{,}730$
t = 11,460 years

Answer: t ≈ 11,460 years

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20.3 A sample initially contains 400 g of a radioactive isotope with a half-life of 10 years. How much remains after 30 years?

Given: $N_0 = 400$ g
$T_{1/2} = 10$ years
$t = 30$ years

To find: $N = ?$

Formula: $n = t / T_{1/2}$, $N = N_0(1/2)^n$

Solution: $n = 30 / 10 = 3$ half-lives
$N = 400 \times (1/2)^3$
$N = 400 / 8$
$N = 50$ g

Answer: $N = 50$ g

20.4 A radioactive substance has a half-life of 64 months. In how much time, one-eighth of the substance will be left?

Given: $T_{1/2} = 64$ months
$N / N_0 = 1/8$

To find: $t = ?$

Formula: $(1/2)^n = 1/8$, $t = n \times T_{1/2}$

Solution: $(1/2)^n = (1/2)^3$, so $n = 3$ half-lives
$t = 3 \times 64$
$t = 192$ months

Answer: $t = 192$ months (16 years)

20.5 In a nuclear reaction, 4 µg of mass is converted into energy. Find the energy.

Given: $m = 4\ \mu g = 4 \times 10^{-9}$ kg
$c = 3 \times 10^8$ m s^{-1}

To find: $E = ?$

Formula: $E = mc^2$

Solution: $E = 4 \times 10^{-9} \times (3 \times 10^8)^2$
$E = 4 \times 10^{-9} \times 9 \times 10^{16}$
$E = 3.6 \times 10^8$ J
$E = 360$ MJ

Answer: $E = 3.6 \times 10^8$ J $= 360$ MJ

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20.6 In 420 days, one-eighth of polonium (Po) remains un-decayed. Calculate the half-life of polonium?

Given: t = 420 days
$N / N_0 = 1/8$
To find: $T_{1/2}$ = ?
Formula: $(1/2)^n = 1/8$, $T_{1/2} = t / n$
Solution: $(1/2)^n = (1/2)^3$, so n = 3 half-lives
$T_{1/2} = 420 / 3$
$T_{1/2} = 140$ days
Answer: $T_{1/2} = 140$ days

20.7 Three-fourth of the initial mass of a certain radioactive isotope decay after one hour. Find half-life of isotope in minutes?

Given: t = 1 hour = 60 minutes
decayed = 3/4, so remaining = 1/4
To find: $T_{1/2}$ = ?
Formula: $(1/2)^n = 1/4$, $T_{1/2} = t / n$
Solution: $(1/2)^n = (1/2)^2$, so n = 2 half-lives
$T_{1/2} = 60 / 2$
$T_{1/2} = 30$ minutes
Answer: $T_{1/2} = 30$ min

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Points to Remember

- A = Z + N: atomic number Z = protons, mass number A = protons + neutrons.
- Alpha: ^4_2He (charge +2); beta: fast electron (charge −1); gamma: high-energy photon.
- Half-life: $N = N_0(1/2)^{t/T}$, $A = \lambda N$, $T_{1/2} = 0.693/\lambda$.
- $E = mc^2$; mass defect appears as energy in fission and fusion.
- Fission → splitting heavy nuclei; fusion → joining light nuclei (the Sun's source).